E17e Fourier Analysis of Coupled Electric Oscillations

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Theoretical Background:

In this experiment, we investigate coupled electrical oscillations in LC-circuits using both capacitive and inductive coupling. The key theoretical concepts and governing equations are summarized below:

1. **Uncoupled Resonant Circuit** A single LC-resonator of inductance L and capacitance C exhibits free (undriven) oscillations whose angular frequency is:

$$\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow f_0 = \frac{\omega_0}{2\pi}$$

These arise from the balance of inductive and capacitive voltages under negligible damping

2. **Capacitive Coupling** Two identical LC-circuits coupled via a capacitor C_K can be arranged in two configurations:
   - **Low-point circuit**: The equations of motion for currents I_A, I_B in each loop are

$$\frac{d^2 I_A}{dt^2} + \frac{1}{LC} I_A + \frac{1}{LC_K}\left(I_A - I_B\right) = 0$$

$$\frac{d^2 I_B}{dt^2} + \frac{1}{LC} I_B + \frac{1}{LC_K}\left(I_B - I_A\right) = 0$$

   By forming $I_{+¿=I_A+I_B¿}$ and $I_- = I_A - I_B$, one finds two eigenmodes:

- **In-phase** ($I_- = 0$) at frequency:

$$\omega_1 = \omega_0 = \sqrt{\frac{1}{LC}}$$

- **Out-of-phase** ($I_{+¿=0¿}$) at frequency:

$$\omega_2 = \frac{1}{\sqrt{L\left(C^{-1} + 2C_K^{-1}\right)^{-1}}}$$

The capacitive coupling factor follows

$$k_{C,T} = \frac{C}{C + C_K}$$

- **High-point circuit** Charging the two capacitors either in-phase ($U_a = U_b$) or anti-phase ($U_a = -U_b$) excites the in- and out-of-phase modes directly. One obtains:

$$\omega_- = \sqrt{\frac{1}{LC}}$$

$$\omega_{+¿=\frac{1}{\sqrt{L(C+2C_K)}}\to k_{C,H}=\frac{C_K}{C+C_k}¿}$$

1. **Beat Oscillations** If only one circuit is excited ($I_A(0)\neq 0, I_B=0$), the superposition of the two modes leads to beats. The beat angular frequency is:

$$\omega_S=\omega_2-\omega_1\to T_S=\frac{2\pi}{\omega_S}$$

1. **Inductive Coupling** Two identical LC-circuits coupled by mutual inductance M obey

$$\frac{d^2 I_A}{dt^2}L+M\frac{d^2 I_B}{dt^2}+\frac{1}{C}I_A=0$$
$$\frac{d^2 I_B}{dt^2}L+M\frac{d^2 I_A}{dt^2}+\frac{1}{C}I_B=0$$

Defining $k_L=\frac{M}{L}$, the eigenfunctions are:

$$\omega_{\pm}=\frac{\omega_0}{\sqrt{1\pm k_L}}$$

and for weak coupling $\Delta\omega=\omega_- - \omega_{+¿\approx k_L\omega_0¿}$.

Experimental Setup and Analysis:

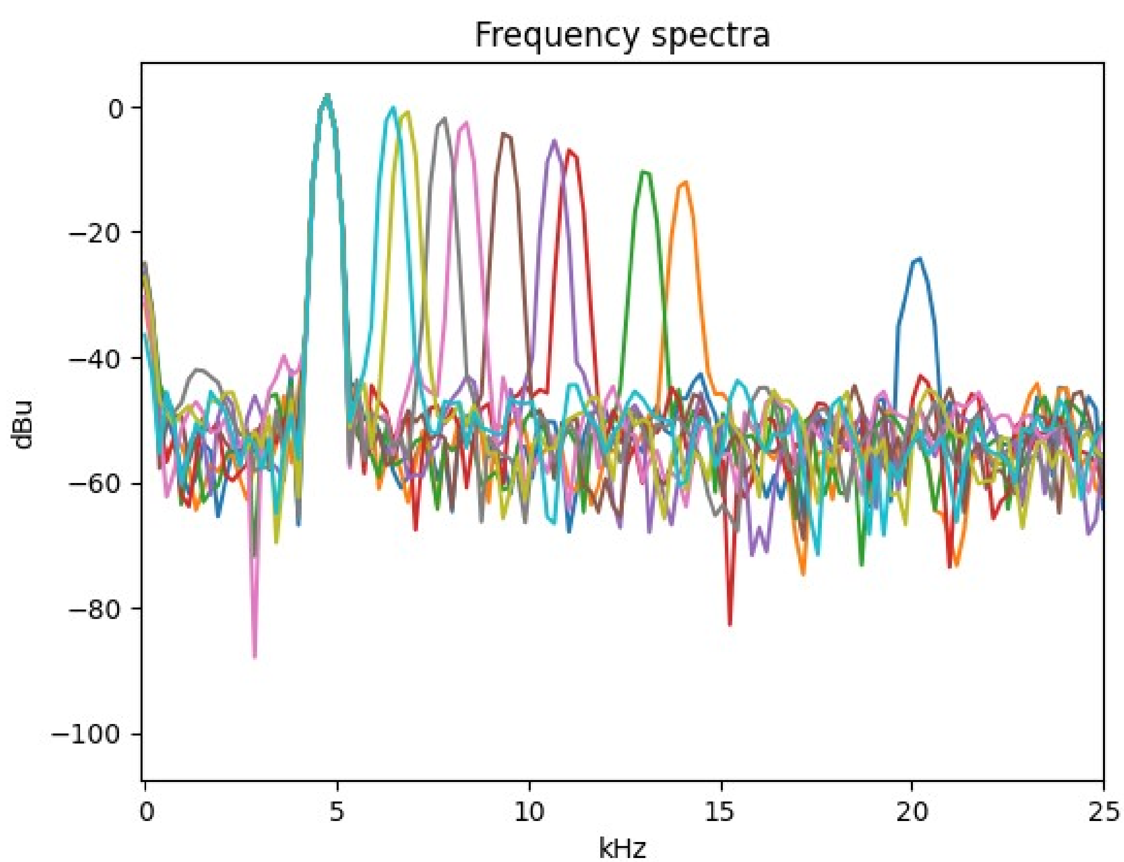
The experiment employs a USB-oscilloscope with FFT capability (PicoScope 2000/5000 series), variable capacitors, coils, resistors, and switches mounted on a circuit board. All measurements are recorded digitally and transferred to a PC for analysis

1. **Low-Point Capacitive Coupling**
   - Assemble the circuit as shown in the figure: two series LC branches in parallel with a variable C_K.
   - Charge capacitors via the DC source U_a, then close the switch to initiate free oscillations.
   - Record time-domain traces and use the FFT function to obtain frequency spectra
   - Repeat for ten values of C_K, exporting all ten spectra for a combined plot.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit

#Values for Ck
Ck = 
np.array([0.002,0.005,0.006,0.009,0.010,0.015,0.020,0.025,0.04,0.05])
Ck = 1.111e-2*Ck

#Data import
A_list = []
for i in range(1,11):
    data1 = pd.read_csv(f'Data/Task1/Task1_freq_{i}.csv', sep=',')
```

```
    freq = data1['Frequency'].to_numpy()
    y = data1['Channel A'].to_numpy()
    A_list.append(y)
A = np.vstack(A_list)

#Filtering the in and out frequencies
rfreq = []
for i in range(0,10):
    x = A[i]>-40
    f = freq[x]
    split = np.where(np.diff(f)>1)[0]
    segments = np.split(f, split+1)
    for j in range(0,len(segments)):
        a = np.mean(segments[j])
        rfreq.append(a)

#in the last measurament the in and out frequencies where so close it
couldnt be differentiated by a simple filter
rfreq.append(rfreq[-1])

rfreq = np.reshape(rfreq,(int(len(rfreq)/3),3))

f_in = rfreq[:,1]
f_out = rfreq[:,2]

#calculations
k = (f_out**2-f_in**2)/(f_out**2+f_in**2)
C = k*Ck/(1-k)

#results for k
print("The valued obtained for k given diferent Ck's is:")
for i in range(0,len(C)):
    print(f'k = {"%.2f"%(k[i])} for Ck = {"%.2f"%(Ck[i]*1e6)}µF')

#results for C
print("The valued obtained for C given diferent Ck's is:")
for i in range(0,len(C)):
    print(f'C = {"%.2f"%(C[i]*1e6)}µF for Ck = {"%.2f"%
(Ck[i]*1e6)}µF')

#plot
plt.figure(1)
for i in range(0,10):
    plt.title('Frequency spectra')
    plt.xlabel('kHz')
    plt.ylabel('dBu')
    plt.xlim(-0.1,25)
    plt.plot(freq, A[i])

plt.show()
```

```
The valued obtained for k given diferent Ck's is:
k = 0.90 for Ck = 22.22µF
k = 0.80 for Ck = 55.55µF
k = 0.77 for Ck = 66.66µF
k = 0.70 for Ck = 99.99µF
k = 0.68 for Ck = 111.10µF
k = 0.61 for Ck = 166.65µF
k = 0.53 for Ck = 222.20µF
k = 0.46 for Ck = 277.75µF
k = 0.35 for Ck = 444.40µF
k = 0.00 for Ck = 555.50µF
The valued obtained for C given diferent Ck's is:
C = 194.90µF for Ck = 22.22µF
C = 222.20µF for Ck = 55.55µF
C = 227.22µF for Ck = 66.66µF
C = 235.05µF for Ck = 99.99µF
C = 234.67µF for Ck = 111.10µF
C = 256.81µF for Ck = 166.65µF
C = 246.94µF for Ck = 222.20µF
C = 240.62µF for Ck = 277.75µF
C = 244.32µF for Ck = 444.40µF
C = 0.00µF for Ck = 555.50µF
```

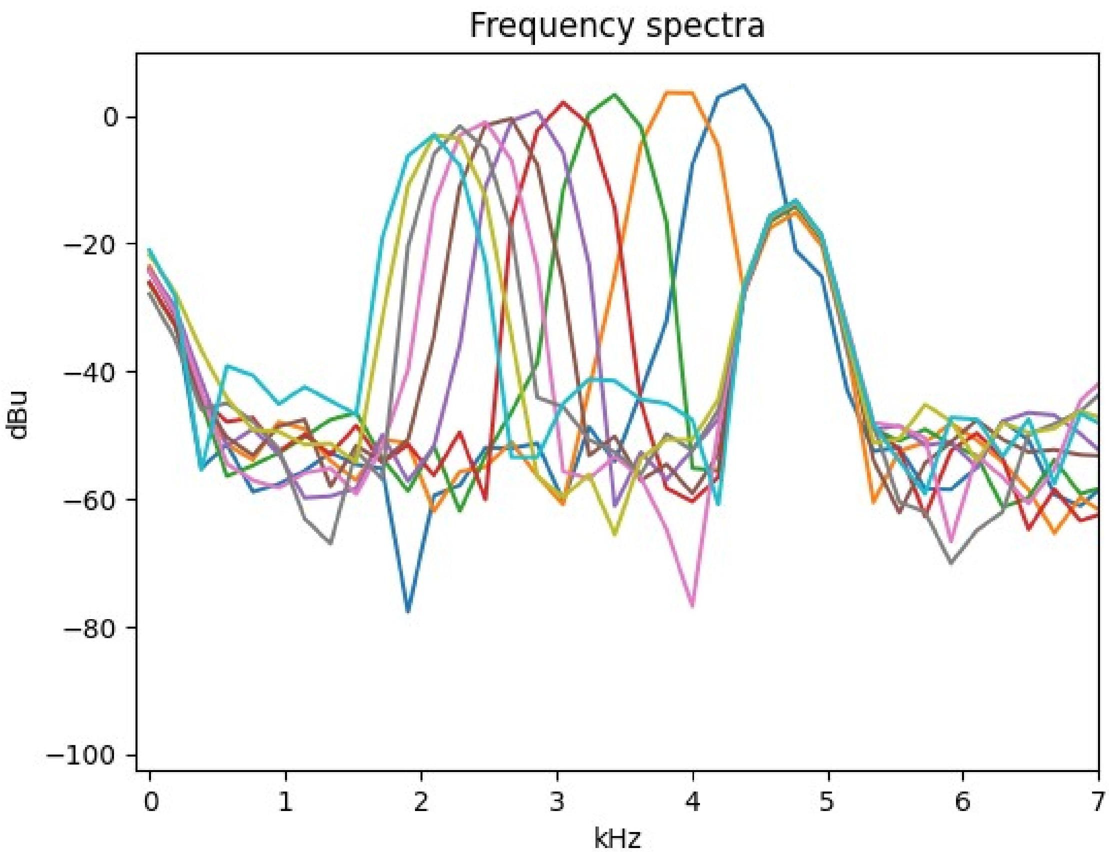


1. **High-Point Capacitive Coupling**
    - Build the high-point circuit as shown in the figure with separate charging switches S_a, S_b.
    - For in-phase mode: set $U_a = U_b$; for out-of-phase: $U_a = -U_b$.
    - Measure and record frequency spectra for ten values of C_K.

```
#Ck values
Ck =
np.array([0.002,0.005,0.01,0.015,0.02,0.025,0.03,0.035,0.04,0.045])

Ck = 1.111e-2*Ck

#data import
A_list = []
for i in range(1,11):
    data1 = pd.read_csv(f'Data/Task2/Task2_freq_{i}.csv', sep=',')
    freq = data1['Frequency'].to_numpy()
    y = data1['Channel A'].to_numpy()
    A_list.append(y)
A = np.vstack(A_list)

#filtering out other frequencies
rfreq = []
for i in range(0,10):
    x = A[i]>-40
    f = freq[x]
    split = np.where(np.diff(f)>0.5)[0]
    segments = np.split(f, split+1)
    for j in range(0,len(segments)):
        a = np.mean(segments[j])
        rfreq.append(a)

#adding noise to the frequencies that where too close together to
differentiate
rfreq = np.insert(rfreq, 2, rfreq[1]+np.random.rand()*1e-3)
rfreq = np.insert(rfreq, 4, rfreq[4]+np.random.rand()*1e-3)

rfreq = np.reshape(rfreq,(int(len(rfreq)/3),3))

#Calculating k and C
f_in = rfreq[:,1]
f_out = rfreq[:,2]

k = (f_out**2-f_in**2)/(f_out**2+f_in**2)

C = (1-k)*Ck/k

#results for k
print("The valued obtained for k given diferent Ck's is:")
for i in range(0,len(C)):
```

```
    print(f'k = {"%.2f"%(k[i])} for Ck = {"%.2f"%(Ck[i]*1e6)}μF')

#results for C
print("The valued obtained for C given diferent Ck's is:")
for i in range(0,len(C)):
    print(f'C = {"%.2f"%(C[i]*1e6)}μF for Ck = {"%.2f"%
(Ck[i]*1e6)}μF')

#plotting
plt.figure(1)
for i in range(0,10):
    plt.title('Frequency spectra')
    plt.xlabel('kHz')
    plt.ylabel('dBu')
    plt.xlim(-0.1,7)
    plt.plot(freq, A[i])

plt.show()
```

```
The valued obtained for k given diferent Ck's is:
k = 0.00 for Ck = 22.22μF
k = -0.00 for Ck = 55.55μF
k = 0.34 for Ck = 111.10μF
k = 0.42 for Ck = 166.65μF
k = 0.50 for Ck = 222.20μF
k = 0.55 for Ck = 277.75μF
k = 0.60 for Ck = 333.30μF
k = 0.63 for Ck = 388.85μF
k = 0.65 for Ck = 444.40μF
k = 0.68 for Ck = 499.95μF
The valued obtained for C given diferent Ck's is:
C = 2696788.26μF for Ck = 22.22μF
C = -581202.19μF for Ck = 55.55μF
C = 213.49μF for Ck = 111.10μF
C = 231.23μF for Ck = 166.65μF
C = 225.28μF for Ck = 222.20μF
C = 228.66μF for Ck = 277.75μF
C = 222.20μF for Ck = 333.30μF
C = 232.82μF for Ck = 388.85μF
C = 238.55μF for Ck = 444.40μF
C = 240.06μF for Ck = 499.95μF
```

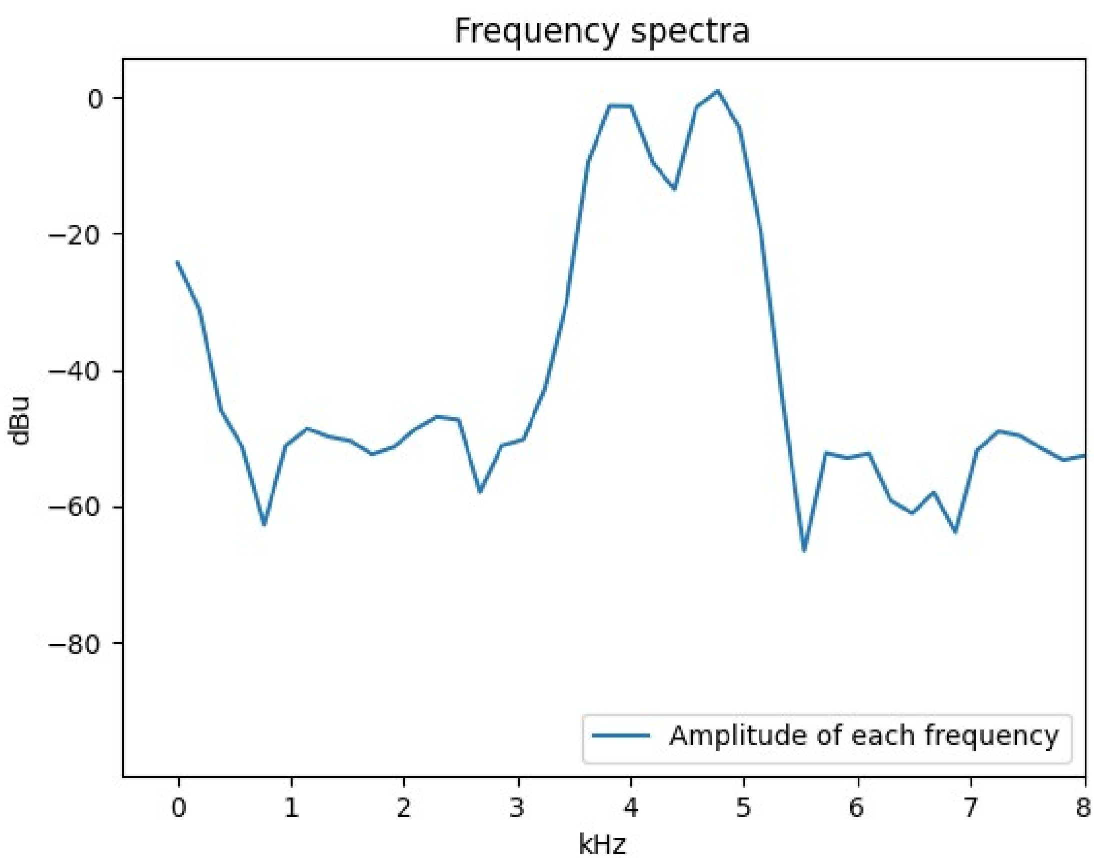


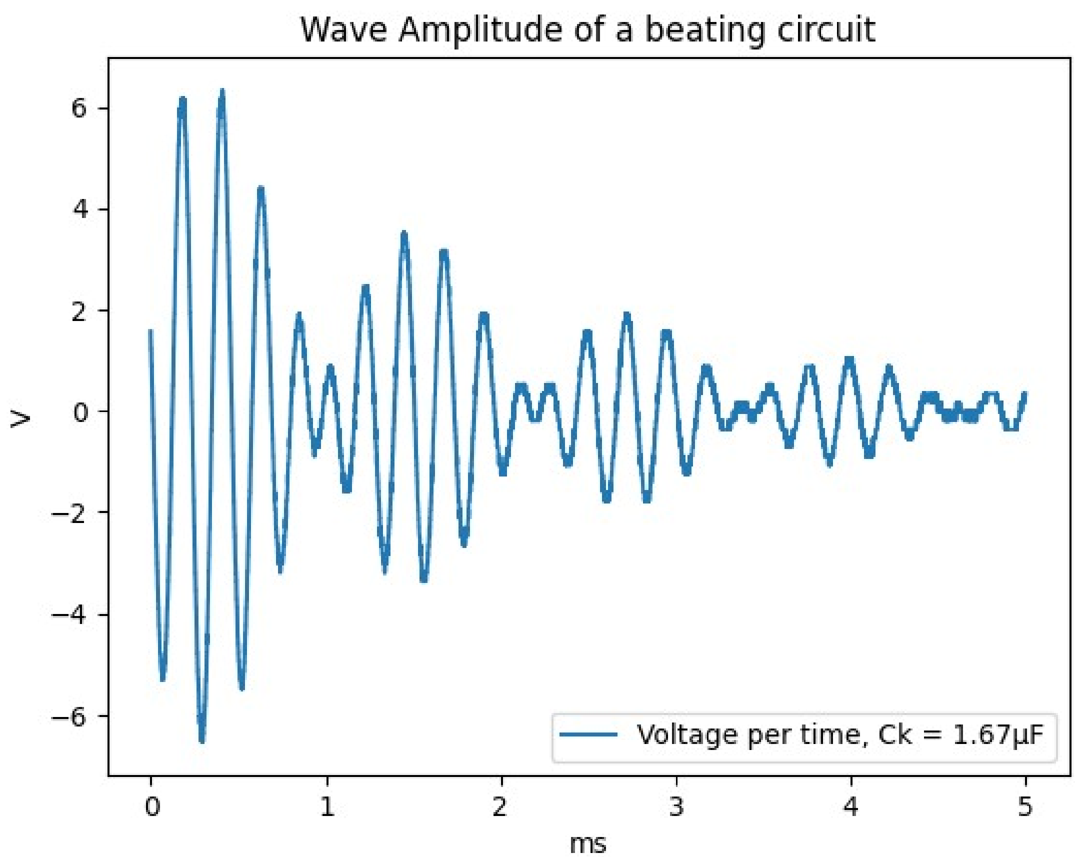
1. **Beat Measurements**
   - In the high-point circuit, short one branch, charge the other, then close the switch.
   - Acquire both time-trace and spectrum of the beat oscillation.

```
#data import
data1 = pd.read_csv('Data/Task3/Task3_freq_1.csv', sep=',')
freq = data1['Frequency'].to_numpy()
Af = data1['Channel A'].to_numpy()

data1 = pd.read_csv('Data/Task3/Task3_A_1.csv', sep=',')

#filtering other frequencies
x = Af > -2
rfrec = freq[x]

#taking the mean values of the peeks
f1 = (rfrec[0]+rfrec[1])/2
f2 = (rfrec[2]+rfrec[3])/2

#calculating the period
T = 2*np.pi/(f2-f1)

#result for T
```

```python
print(f'The period of the beating is: {"%.2f"%T} ms')

#plotting
plt.figure(1)
plt.xlim(-0.5,8)
plt.title('Frequency spectra')
plt.xlabel('kHz')
plt.ylabel('dBu')
plt.plot(freq,Af, label='Amplitude of each frequency')
plt.legend(loc = 'lower right')

plt.figure(2)
plt.title('Wave Amplitude of a beating circuit')
plt.xlabel('ms')
plt.ylabel('V')
plt.plot(data1['Time'], data1['Channel A'], label=f'Voltage per time, Ck = {"%.2f"%(0.015*1.111e2)}μF')
plt.legend(loc='lower right')
plt.show()
```

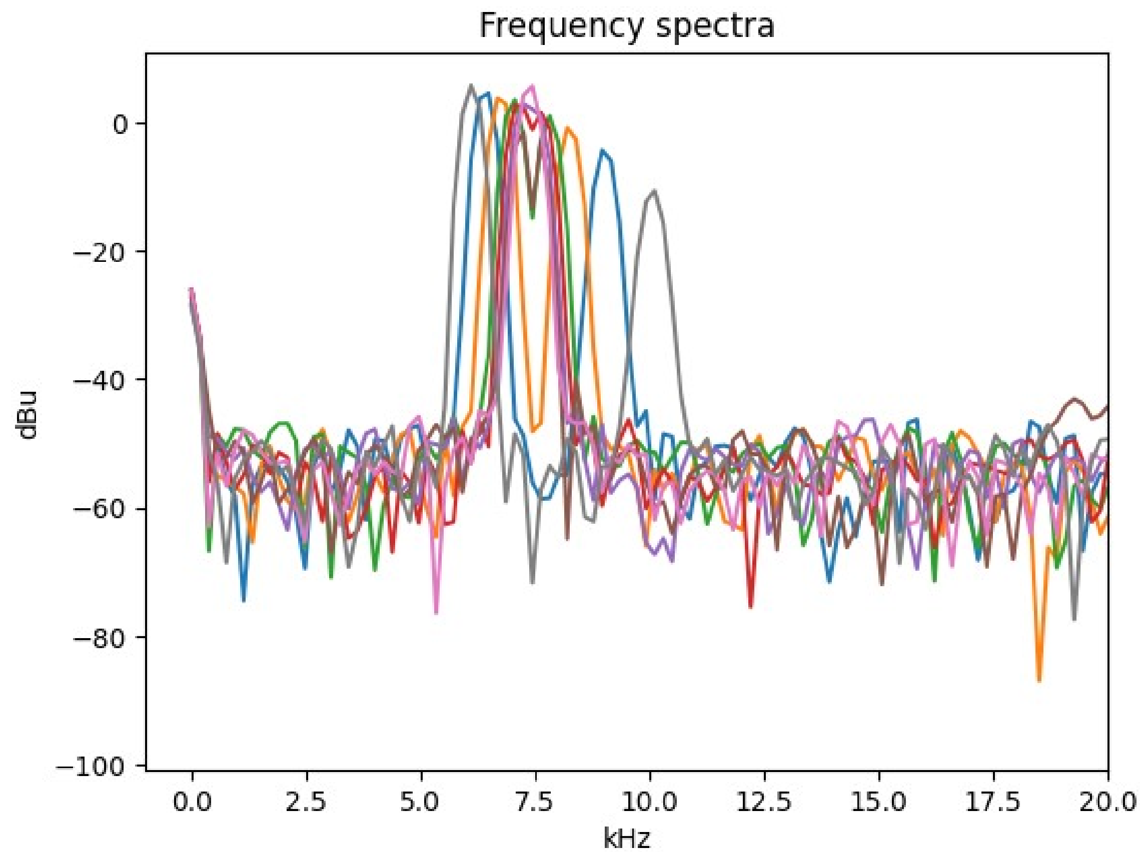
```
The period of the beating is: 8.24 ms
```





1. **Inductive Coupling**
   - Use two identical flat coils facing each other.
   - ary the coil separation d over eight positions.
   - For each d, measure uncoupled f_0 and coupled $f_\pm$ to compute M and K_L.

```
A_list = []

for i in range(1,9):
    data1 = pd.read_csv(f'Data/Task4/Task4_freq_{i}.csv', sep=',')
    freq = data1['Frequency'].to_numpy()
    y = data1['Channel A'].to_numpy()
    A_list.append(y)

A = np.vstack(A_list)

for i in range(0,8):
    plt.title('Frequency spectra')
    plt.xlabel('kHz')
    plt.ylabel('dBu')
    plt.xlim(-1,20)
    plt.plot(freq, A[i])
```



All instrument settings (time base, trigger level, FFT parameters) are optimized to resolve both f_1 and f_2 clearly. Data are logged systematically for subsequent plotting and fitting.